QABG Paper Template

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This is the abstract section. You should provide a concise summary of your paper here. Describe the main problem, your approach, key contributions, and results. The abstract should be self-contained and give readers a clear understanding of what your paper is about.

1. Introduction

This is the introduction section where you should motivate your work and provide necessary background. Start by explaining the problem you are addressing and why it is important. Then discuss existing approaches and their limitations. Finally, briefly introduce your method and summarize your main contributions.

You can organize the introduction as follows:

- Background and motivation
- Problem statement
- Limitations of existing methods
- Overview of your approach
- Summary of contributions

Make sure to cite relevant work throughout the introduction using the `\citep{}` or `\citet{}` commands.

2. Preliminary

In this section, you should introduce the necessary background knowledge and formal definitions that readers need to understand your work. This might include:

- Mathematical notation and definitions
- Background on the problem domain
- Description of the framework or methodology you build upon

Make sure to define all symbols and notation clearly. For example, if you use $\boldsymbol{x}$ and $\boldsymbol{y}$ to denote input and output, define them explicitly.

3. Methodology

This section describes your proposed method in detail. You should:

1. Provide a high-level overview of your approach
2. Describe each component of your method
3. Explain the theoretical justification (if applicable)
4. Discuss implementation details

You may want to include:

- Mathematical formulations
- Algorithm pseudocode (using the algorithm2e package)
- Figures illustrating your approach

For example, you can present your objective function:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\mathcal{L}(f_\theta(x), y)\right] \tag{1}$$

where $\mathcal{L}$ is the loss function and f_θ is your model.

4. Related Work

In this section, you should review relevant prior work and position your contributions relative to existing literature. Organize this section by topics or themes rather than simply listing papers.

Key aspects to cover:

- Major approaches in the field
- Strengths and limitations of existing methods
- How your work differs from or builds upon prior work

Make sure to cite all relevant papers and provide a clear comparison between your approach and existing methods.

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5. Experiments

This section presents your experimental evaluation. Include the following subsections:

5.1. Experimental Setup

Describe your experimental setup, including:

- Datasets used
- Baseline methods for comparison
- Evaluation metrics
- Implementation details and hyperparameters
- Hardware and computational resources

5.2. Main Results

Present your main experimental results here. Use tables and figures to illustrate the performance of your method compared to baselines.

You can create tables like this:

Table 1 | Comparison with baseline methods

Method	Metric 1	Metric 2
Baseline 1	85.2	78.3
Baseline 2	87.1	80.5
Ours	**92.3**	**86.7**

5.3. Analysis and Ablation Studies

Provide additional analysis to understand why your method works:

- Ablation studies to evaluate the contribution of different components
- Sensitivity analysis for hyperparameters
- Qualitative examples or case studies

6. Conclusion

In the conclusion, summarize your main contributions and findings. Discuss the implications of your work and potential future directions.

Structure your conclusion as follows:

- Brief recap of the problem and your approach
- Summary of key results and contributions
- Limitations of your work
- Future research directions

Keep the conclusion concise and avoid introducing new information.

A. Additional Details

You can include supplementary material in the appendix, such as:

- Detailed proofs
- Additional experimental results
- Implementation details
- More examples or visualizations

B. Proofs

Include detailed mathematical proofs here if needed.

$$
\begin{aligned}
I(\boldsymbol{x};\boldsymbol{y}) &= H(\boldsymbol{y}) + \mathbb{E}_{p(\boldsymbol{x},\boldsymbol{y})}[\log p(\boldsymbol{y}|\boldsymbol{x})] && (2)\\
&\geq H(\boldsymbol{y}) + \mathbb{E}_{p(\boldsymbol{x},\boldsymbol{y})}[\log q_\theta(\boldsymbol{y}|\boldsymbol{x})] && (3)
\end{aligned}
$$